Coding & Solution

Date	03 July 2026
Team ID	XXXXX
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Coding & Solution

This section evaluates the quality and completeness of your implemented code and the overall solution. Your submission should demonstrate working functionality, clean code practices, and alignment with your defined requirements.

Instructions:

1. Ensure your code is well-structured, commented, and follows best practices for your chosen technology stack.
2. The solution should address the core problem statement and implement the key functional requirements.
3. Include all source files, configuration files, and setup instructions needed to run the application.
4. Attach or link to your code repository (e.g., GitHub) in the table below.

Solution Summary

Field	Details
Repository Link / URL	https://github.com/varalakshmivennapusa124-cell/OptiCrop.git
Programming Language(s)	Python, HTML5, CSS3, JavaScript
Framework(s) Used	Flask, Scikit-learn, Pandas, NumPy, Matplotlib
Key Features Implemented	• Soil and environmental data input (N, P, K, Temperature, Humidity, pH, and Rainfall). • Machine learning–based crop recommendation using agricultural data. • Crop suitability analysis with suitability percentage and parameter gap analysis. • Top 3 crop recommendations with confidence scores and farming tips. • Responsive web interface with Flask API integration for real-time predictions.
Pending / Incomplete Features	• Real-time weather API integration •Fertilizer recommendation module •Crop disease prediction •Mobile application support
Setup / Run Instructions	install the required dependencies,run the flask backend using python app.py and open the application in a browser using the local flask server

Code Quality Checklist

S.No	Criteria	Status
1	Code is modular and organized into functions / classes	Yes
2	Meaningful variable and function names are used	Yes
3	Code includes comments / documentation where necessary	Yes
4	Error handling is implemented for critical operations	Yes
5	The application runs without critical errors	Yes
6	Code is committed to a version control repository	Yes

Additional Notes / Comments:
The OptiCrop project is designedto help farmers make better crop selection decisions by analyzing soil nutrients and environmental conditions. The application follows a simple and organized structure, making it easy to understand, maintain, and improve. Machine learning is used to generate crop recommendations, while Flask connects the prediction model with a user-friendly web interface. The project can be extended in the future by adding live weather integration, fertilizer recommendations, and crop disease detection.